

YITIAN WANG



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SUMMARY

Ph.D. in Electrical Engineering with expertise in thin-film deposition (PVD/CVD), crystal growth, materials characterization, and process optimization. Experienced in DOE-based process development, hardware troubleshooting, and large-scale data analysis using Python and MATLAB. Led multi-institutional experiments at Oak Ridge National Laboratory and published 5 first-author papers. Strong foundation in semiconductor device physics and CMOS process flows through graduate-level coursework.

EDUCATION

University of California, Riverside, CA *Mar. 2021 - Jun. 2026 (expected)*

Ph.D. in Electrical and Computer Engineering

Awards: MRS 2023 Highly Commended Student Talk Award; Dissertation Completion Fellowship

Coursework: Semiconductor Device Physics, CMOS Fabrication, Thin-Film Technology

Columbia University in the City of New York, NY *Aug. 2019 - Feb. 2021*

Master of Science, Material Science and Engineering

Beijing Normal University, CN *Sep. 2015 - Jun. 2019*

Bachelor of Science, Physics

Awards: 'Jingshi' Scholarship (2016 - 2018)

SKILLS

Process & Deposition	PVD / Magnetron Sputtering, CVD, Floating-Zone Crystal Growth, DOE
Characterization	SEM, TEM, XRD, AFM, Raman, FTIR, Inelastic Neutron Scattering, PPMS, DSC, TGA, Four-Probe Transport
Software	Python, MATLAB, PyTorch, R, SQL, GSAS-II, Mantid, ImageJ, Solid-Works, Quantum ESPRESSO, VASP, COMSOL
Electrochemistry	Prototype Cell Assembly, Separator Fabrication, EIS, Galvanostatic Cycling

RESEARCH AND ENGINEERING EXPERIENCE

Graduate Research Assistant — Process Development & Characterization *Jun. 2021 - Jun. 2025*

University of California, Riverside

Riverside, CA

- Designed and executed DOE-based experiments to optimize crystal growth and thin-film deposition processes for solid-state electrolyte materials.
- Performed systematic hardware characterization on floating-zone furnaces and deposition systems, troubleshooting process issues and implementing parameter improvements.
- Measured and analyzed film and bulk material properties (XRD, SEM, Raman, FTIR, four-probe transport) to establish process-property correlations.
- Built automated data analysis and fitting frameworks in MATLAB and Python, generating technical reports and presentations for cross-functional teams.

- Managed vendor relationships (Thermo Fisher, Sigma-Aldrich, MTI) and mentored junior graduate students on experimental techniques.

Experiment Lead — Neutron Scattering Characterization

Jan. 2022 - Jul. 2023

Oak Ridge National Laboratory

Oak Ridge, TN

- Led complex characterization experiments on solid electrolytes using TAX (HFIR) and ARCS (SNS) instruments, coordinating with beamline scientists to optimize experimental parameters.
- Developed automated data processing and analysis workflows in Mantid and Python, reducing analysis turnaround and improving reproducibility.
- Generated technical documentation and coordinated across 5+ institutions including ORNL, ETH Zürich, and TU Wien.

Graduate Researcher — Battery Materials & Process Engineering

Sep. 2019 - Jan. 2021

Columbia University

New York, NY

- Developed composite separator fabrication processes using solution processing of γ -C₃N₄/PVDF, optimizing film uniformity and mechanical properties.
- Assembled prototype coin cells and performed electrochemical characterization (EIS, galvanostatic cycling) to evaluate process quality and device performance.
- Conducted first-principles simulations (Quantum ESPRESSO) to predict defect behavior and guide process design for olivine-type electrode materials.

Student Researcher — Hardware Design & Characterization

Sep. 2018 - May 2019

Beijing Normal University

Beijing, CN

- Designed precision uniaxial-strain devices in SolidWorks for thin-film sample characterization.
- Performed systematic transport measurements using PPMS, developing testing protocols for temperature-dependent material studies.

PROFESSIONAL MEMBERSHIPS

Sigma Xi (Elected Full Member, 2026) · American Physical Society (2025–2026) · Materials Research Society (through 2027)

SELECTED PUBLICATIONS (5 FIRST-AUTHOR)

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1. **Y. Wang** et al. “Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte.” *PRX Energy*, 2025. [\[Link\]](#)
 2. **Y. Wang** et al. “Glass-like thermal transport in perovskite Li-ion conductor.” *Chem. Commun.*, 2025 (Front Cover). [\[Link\]](#)
 3. **Y. Wang** et al. “Low thermal conductivity of NaSICON-type solid electrolyte.” *Tungsten*, 2025. [\[Link\]](#)
 4. **Y. Wang** et al. “Thermal properties of Li-ion conducting garnet solid electrolyte.” *J. Mater. Chem. A*, 2024. [\[Link\]](#)
 5. **Y. Wang**, X. Chen. “Single crystal growth of garnet-type fast Li-ion conductors.” *Tungsten*, 2022. [\[Link\]](#)